

Insecure Mail Puzzle



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- ② The mail is corrupt!: Unlocked packages will be emptied!
- ③ Locked packages will be left alone by lazy official.
- ④ No one can open a locked package without the key.
- ⑤ How do you send?!



Insecure Mail Puzzle



- ① One way is for Mr. Rogers and Ms. Aberlin each to start with *their own* locks...



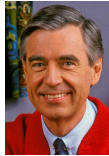
Insecure Mail Puzzle



- ① One way is for Mr. Rogers and Ms. Aberlin each to start with *their own* locks...
- ② R locks box with his lock



Insecure Mail Puzzle



- ① One way is for Mr. Rogers and Ms. Aberlin each to start with *their own* locks...
- ② R locks box with his lock and mails...



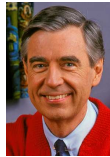
Insecure Mail Puzzle



- ① One way is for Mr. Rogers and Ms. Aberlin each to start with *their own* locks...
- ② R locks box with his lock and mails...
- ③ A *also locks* box with her lock



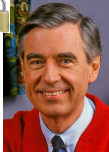
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- ① One way is for Mr. Rogers and Ms. Aberlin each to start with *their own* locks...
- ② R locks box with his lock and mails...
- ③ A *also locks* box with her lock and mails back...



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- ② R locks box with his lock and mails...
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- ④ R unlocks his lock



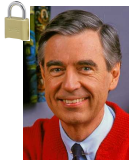
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- ④ R unlocks his lock and mails again...



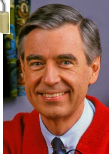
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- ④ R unlocks his lock and mails again...
- ⑤ A unlocks her lock



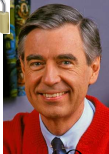
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- ③ A *also locks* box with her lock and mails back...
- ④ R unlocks his lock and mails again...
- ⑤ A unlocks her lock and now she has the gift!



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- ④ R unlocks his lock and mails again...
- ⑤ A unlocks her lock and now she has the gift!
- ⑥ The Diffie-Hellman Protocol in essence does this through algebra (and a complexity assumption)!

